R.D. DUTTA & CO. PVT. LTD.

302, Rattan Jyoti Building 18,
Rajendra Place, New Delhi - 110008
Phone : 25710739, 25739805
Email : info@rddutta.com
Web : www.rddutta.com
CIN No. : U74899DL1982PTC014209

Quotation No : Q24000494

Date : 19-Sep-2024

Chanderpur Works Pvt. Ltd.
Yamunanagar
YAMUNA NAGAR
HARYANA

Kind Attention : Mr./Ms. Akshay Chandra

Dear Sir,

It's been more than 30 years since R D Dutta & Company came into existence in the field of Compressed air systems and accessories.

We have an expertise in Providing best of sales and after sales support and a wide range of products to meet our customer needs.

To us being leader not only means selling the best quality products but, it also means working hard to understand your industry, your needs and the demands placed on your productivity in order to provide you with the most beneficial solutions for your work.
We trust you Find our offer in line with your requirement.

Should you require any further clarification/information, please contact the undersigned.
Thanking you of out best attention at all times.

Very truly yours,

For R. D. Dutta & Co. PVT. LTD.

Nakul Awasthy

Sales Head

9311955933

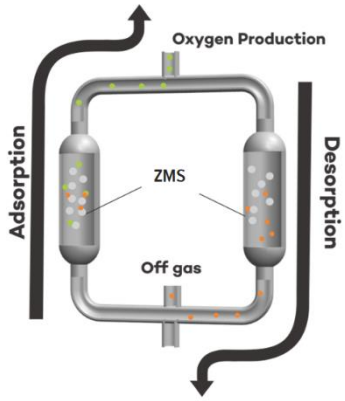
DT1005

idos customs

Industrial PSA Oxygen Generator



PSA Technology



PSA Process explaining oxygen production

Industrial Oxygen Generator mainly consists of two adsorbers filled with zeolite molecular sieves. Pressurized air is passed through one of these adsorbers, zeolite having a property to attract nitrogen, adsorbs Nitrogen from air and oxygen is allowed to pass through and move into the storage tank as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of Oxygen. This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of oxygen produced is monitored by an Oxygen analyser.

Key Features

- Shipped ready for installation
- 5 minutes to run and never have to stop it!
- Easy installation
- Low operating expenses compared to gas cylinders/liquid Oxygen
- Maintenance costs are very less
- Embedded Safety Interlocks
- Space-saving design
- Eliminates all hazards which are there with gas cylinder or liquid Oxygen
- Longevity: at least 10 years

absstem REINVENTING PERFORMANCE

Absstem Industrial Solutions Pvt. Ltd.
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

Description

a.	Product Name	Oxygen Gas Generator
b.	Model Range	**IDOS C**
c.	Technology	PSA (Pressure Swing Adsorption)

Specifications

a.	Oxygen Flow Rate	As per Performance Sheet
b.	Oxygen Purity	As per Performance Sheet
c.	Oxygen Delivery Pressure	5.5 Barg (Higher Pressure Options Available)
d.	Ambient conditions	
	Temperature	0 to 50 Deg. Celsius
	Relative Humidity	up to 80 % (Max.)
	Altitude	up to 1000 m above Sea level

Scheme Offered – (PSA Only)

This scheme will comprise of

- Air package with inlet air filtration system
  (Air compressor, Air dryer, Water Separator, Oil filters)
- Industrial Oxygen generator
- Storage Tank
- Oxygen Analyzer

Unique Features of Absstem

a. All the fabricated items are cleaned and degreased following our strict quality policy.
b. All the Instruments are specially manufactured to make them suitable for Oxygen gas.
c. Guaranteed 10+ years of healthy life.
d. We provide the purest **Oxygen** in Industry.
e. Data logging and Graphical Reports for all parameters available.
f. Test certificates of all instruments are available on demand.

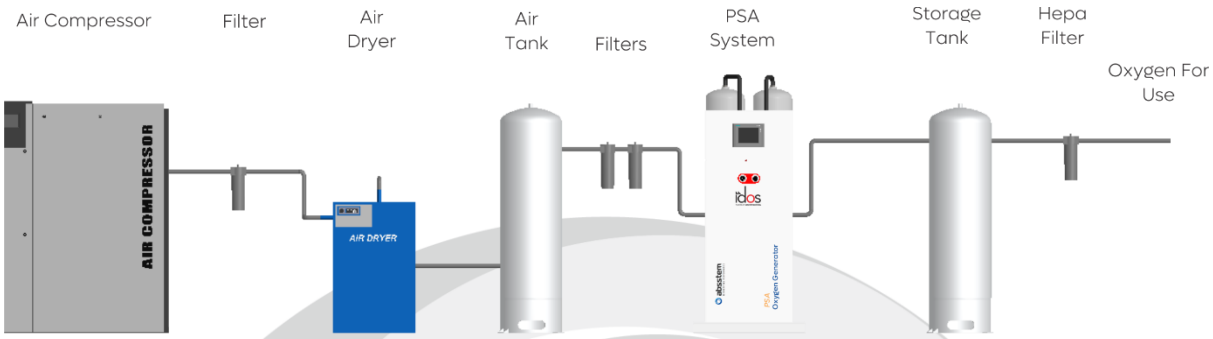
Onsite Gas Generation vs Liquid Tank/Cylinder Alternative

Non-stop availability: 24 hours a day, 7 days a week
Easy integration within existing installations
Your own independent supply of industrial gas
Significant economies for scale and lower operational costs: no rental charges, logistical expenses and bulk user evaporation losses
No safety hazards when handling high-pressure cylinders

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Equipment Description

The Industrial Oxygen Generator consists of

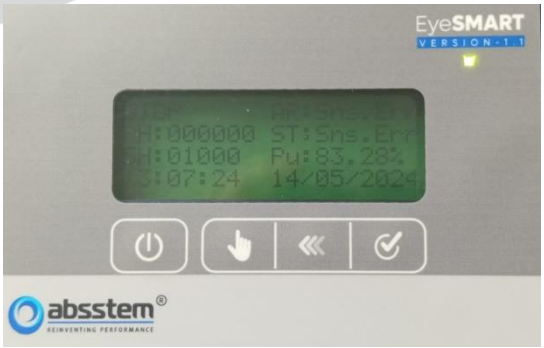


Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Oxygen** from the air.
Refrigeration dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Tank	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Oxygen** from Air.
Storage Tank	Industrial Grade **Oxygen** produced from IDOS C is stored in this tank to supply a continuous flow of **Oxygen**.
Oxygen Analyzer	Oxygen produced is monitored by an **Oxygen** analyser. In case, O2 purity goes below the specified level, it gives an alarm and switches it to vent mode.

EYE-SMART Touch Screen Display

(User-friendly touchscreen interface for easy operation and maintenance)

Absstem industrial Oxygen generator complies with all industry standards and regulation



- Alarm management system
- Plug and play
- Fully automatic
- Process data display
- Consumables running hours display
- Data logging via USB drive
- Graphical representation

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Additional upgrades

- Digital flow meter with totaliser
- High-pressure Oxygen tank
- Cylinder manifold and alarm management system
- Remote monitoring

Plant Operation

- Unit operates 24-hrs a day, 7-days a week reliably, automatically & safely

Automation

- The Plant is fully automatic and needs only electrical power for running the Air compressor
- The machine automatically enters into standby mode when the level in the Storage Tank reaches its set limit. This helps to increase the life of the Air compressor.

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Oxygen generators.

Itemised Supply description

No.	Item description	Make	Qty
a	Air Compressor with Filters	ABSSTEM APPROVED	Client Scope / Quoted Separately
b	Refrigeration Dryer	ABSSTEM APPROVED	Client Scope / Quoted Separately
c	Air Tank	ABSSTEM	Client Scope / Quoted Separately
d	Set of Filters	ABSSTEM	1 no.
e	PSA unit (IDOS C)	ABSSTEM	1 no.
f	Electrical Panel	ABSSTEM	1 no.
f	Digital Oxygen Analyzer	ABSSTEM	1 no.
g	Storage Tank	ABSSTEM	1 no.

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Electrical Protection

The client should make proper arrangements for earthing/grounding systems for the Industrial **Oxygen** Generator at their facility.

Enclosure Panel

- Ergonomically designed metal enclosure with powder-coated finish for longer life.
- High-strength IP-rated panel fittings from Germany for a robust enclosure.
- Fan filters from **Rittal** are IP-55 rated making the electrical enclosure's internal environment dust-free.
- Easy connectivity to any place of installation with req. port connections.
- High-strength base frame designed for higher load and easy transportation.
- Vessels are designed per ASME section VIII DIV I. Vessels are epoxy paint coated which is considered good for longevity and corrosion resistance.
- All electrical items are CE certified with operating voltage at 24 VDC

Annual Maintenance Contract (AMC)

Our in-house support Absstem shield is 24*7 open and we assure a timely service maintenance schedule. Per the schedule mentioned in the operational manual, we suggest the operator change the filters by following a simple protocol.

It is very easy to change or replace as our commissioning engineer plus our commissioning engineer and tech support office will always guide you about the activity.

Best ROI Guaranteed

ROI within 6 to 18 months when compared to conventional methods

Generator performance depends on the component selection. Our processes make this job easier as all equipment is selected based on more than 3 decades of research. IDOS C is a fully automatic (operator-free operation) that ensures peace of mind.

We deliver what we commit!!

Thank You

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Product Description

Product	Description	Make	Qty
IDOS Air Package	IDOS 15 CP Oxygen		1

Caption	Specification
Air Compressor KW	18 KW
Air Compressor Make	Atlas Copco/ IR/ Power Efficient Brands
Air Dryer	Included
Inter Connecting Cable	Included
Interconnecting Piping	Included
Pre filter	Included
Dry Air Generated	115 CFM

Product	Description	Make	Qty
IDOS C	IDOS 15 C Oxygen		1

Caption	Specification
Gas Generated	Oxygen
Flow	15 Nm3/Hr
Purity	95 Percent
Compressor	In Your Scope / Quoted Separately
Dryer	In Your Scope / Part of Compressor Package
Air Vessel	In Your Scope / Part of Compressor Package
Dry Air Required	115 CFM
Power Required	230V 50 Hz Single Phase
Module Option	PSA Plant Only

Product	Description	Make	Qty
IDOS Gas Storage	IDOS 15 ST Oxygen		1

Caption	Specification
Water Volume	3000 Litres
Gas Stored	Oxygen
Gas Stored Volume	16.5 M3
Working Pressure	5.5 Barg
Hydrotest Pressure	9 Barg

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Product	Description	Make	Qty
LP Booster 40 Barg	MOC 15 LP 40 Barg Booster	ABSSTEM	1

Caption	Specification
Suction Pressure	4 to 5 Barg
Discharge Pressure	35 to 40 Barg
Voltage	415 VAC
Flow	15 Nm3/hr
Type	Air Cooled Oil Free Reciprocating Booster
Input Power	5.5 KW
Gas to be Compressed	Oxygen / Nitrogen

Product	Description	Make	Qty
HP Storage Tank 40 Barg	1000 ltr HP Storage Tank 40 Barg High Pressure Storage Tank	ABSSTEM	1

Caption	Specification
Capacity	1000 litres
Working Pressure	40 Barg
Hydrotest Pressure	66 Barg
Shell / Dish Thickness	18 mm / 20 mm
In / Out Connection	1 1/2" BSPF
Design Standard	IS 2825

Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	GST(%)	Amount
1	IDOS 15 CP	IDOS 15 CP	IDOS Air Package	8414	1	11,37,600.00	18.00	11,37,600.00
2	IDOS 15 C	IDOS 15 C	IDOS C	842139	1	15,72,000.00	18.00	15,72,000.00
3	IDOS 15 ST	IDOS 15 ST	IDOS Gas Storage	842139	1	2,52,000.00	18.00	2,52,000.00
4	BOS4	MOC 15 LP 40 Barg	LP Booster 40 Barg	84144030	1	8,19,000.00	18.00	8,19,000.00
5	HPS10	1000 ltr HP Storage Tank 40 Barg	HP Storage Tank 40 Barg	73110090	1	2,25,600.00	18.00	2,25,600.00

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Sub Total	40,06,200.00
GST(%) @18.00	7,21,116.00
Grand Total	47,27,316.00

Terms And Conditions

1.	EXCLUSIONS & CLIENTS SCOPE	All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required).
2.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
3.	P&F	3% Extra.
4.	Freight	Extra at Actuals.
5.	Insurance	In your Scope.
6.	Payment Terms	50% Advance with your Prestegious Purchase Order and Balance against PI.
7.	Delivery	6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability.
8.	Warranty	We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
9.	GST	As per the Commercial Table.
10.	Remarks	.

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For R. D. Dutta & Co. PVT. LTD.

Nakul Awasthy

Sales Head

9311955933

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